

Introduction for students: why Chapter 7 matters

Chapter 7 puts together everything in the course. When a switch moves in a circuit with one inductor or one capacitor, every current and voltage moves from an initial value to a final value along an exponential curve. Chapter 7 runs through weeks 12 and 13, and its quiz is at the start of week 14. The final exam covers Chapters 4 to 7.

What you will learn

- **The natural response:** stored energy drains through resistors, and the inductor current or capacitor voltage decays as $\exp(-t/\tau)$.
- **The step response:** a dc source drives the inductor current or capacitor voltage from its initial value to a new final value.
- **The time constant:** $\tau = L/R$ for an inductor and $\tau = RC$ for a capacitor, where R is the Thévenin resistance the element sees after switching. After one τ , 37% of the change is still to come; after five, less than 1%.
- **One formula for both:** $x(t) = x_f + (x_0 - x_f)\exp(-t/\tau)$. Find the initial value, the final value and τ , and the answer follows.
- **What cannot jump:** the inductor current and the capacitor voltage are the same just after switching as just before. Resistor currents and voltages can jump, unless the circuit forces them to follow i_L or v_C .
- **Dependent sources:** the resistance that sets τ then needs a test source, as in Chapter 4.

Where the earlier chapters come back

Chapter	What Chapter 7 uses from it
Chapters 2 and 3: Kirchhoff's laws, dividers	The circuit before and after switching is a resistive circuit
Chapter 4: Thévenin equivalents	τ uses the Thévenin resistance the element sees; dependent sources need a test source
Chapter 6: Inductors and capacitors	The continuity rules give the initial value, and the dc rules (short circuit, open circuit) give the final value

In your program

- **Electrical and Electronics:** motor starting currents, relay and contactor timing, and the charging of the capacitors in every power supply are first-order responses.
- **Computer, Communication and Telecom:** every logic signal charges the capacitance of the next gate through a resistance; that RC time constant limits how fast a chip can run.
- **Biomedical:** in Nilsson's simplified model, a pacemaker times each heartbeat with one resistor and one capacitor (S14, and Nilsson's Practical Perspective for this chapter). Real pacemakers use a crystal-timed microcontroller, but the RC idea is the same.
- **Mechanical and Industrial:** solenoid valves and relays take a few milliseconds to pull in and let go, set by L/R (S15).
- **Surveying:** sensors are read after their RC input filters have settled; waiting less than about five time constants gives a wrong reading.

As an engineer

- **Three numbers decide the answer:** the initial value, the final value and τ . Find each one from its own circuit: before switching, long after switching, and with the independent sources off.
- **Use the right variable:** solve for the inductor current or the capacitor voltage first, because only these are guaranteed to be continuous. Get every other quantity from it.
- **Polarity first:** an initial voltage or current that points against the reference is negative. Read the figure before you write $x(0)$.

In real life

- **A pacemaker (Nilsson's simplified model):** a capacitor charges toward the battery voltage; at 75% of it, a controller fires a pulse into the heart and starts again. Doubling R halves the heart rate (S14).
- **A relay letting go:** after the switch opens, the coil current decays through a diode; adding a resistor shortens τ , so the relay releases sooner, but raises the voltage spike (S15).

How to study this chapter, and where AI fits

- AI tools are allowed on the practice sheet. Use them to check your work, not to replace it.
- In a 2025 study of Gemini 2.5 Pro on undergraduate circuit problems, misread source polarities caused 6 of its 17 wrong answers, and misread current directions 5 (a small sample, but a clear pattern; arXiv 2512.10159)¹. In this chapter they show up as an initial voltage with the wrong sign, or a time constant built from the wrong resistance. S11 shows a polarity slip, and S12 a wrong resistance: find them before you trust any answer.
- Check every answer yourself: $x(0)$ must equal the initial value, $x(\infty)$ the final value, and τ must come from the circuit after switching.
- The quiz is device-free, and every question is a twin of a practice-sheet question. If you can do the sheet on your own, you will do well on the quiz.

Checklist: you can do these without help

- ☐ Find the initial inductor current or capacitor voltage from the circuit before switching
- ☐ Find the final value from the circuit long after switching
- ☐ Find τ from the resistance the element sees, with a test source when there is a dependent source
- ☐ Write $x(t) = x_f + (x_0 - x_f)\exp(-t/\tau)$ and find other currents and voltages from it
- ☐ Say which quantities can jump at $t = 0$ and find their values at $t = 0^+$
- ☐ Work backwards from a measured response to τ , R , L or C

¹<https://arxiv.org/html/2512.10159>